Deepan Chandrasekaran

2300032731cse3@gmail.com | +91 9384331251 | Karur, India | linkedin.com/in/deepanmpc | https://github.com/deepanmpc

PROFESSIONAL SUMMARY

Innovative developer with deep practical experience in modern AI and backend architectures, including RAG systems, CNN-based gesture recognition, and real-time speech pipelines. Skilled in the full stack from React and TypeScript to Python and ChromaDB. Committed to building accessible, high-performance software, currently leveraging microservices and containerisation to solve complex real-world challenges.

EDUCATION

B.Tech in Computer Science and Engineering at KL University

Expected: April 2027 | Vijayawada

CGPA: 9.15 / 10. Coursework: DSA, Operating Systems, DBMS, OOP, Computer Networks, Software Engineering

WORK EXPERIENCE

Website Designer at Local Apparel Manufacturer

2023 – 2023 | Freelance

- Built a full-stack shirt customization website with responsive frontend and order management backend, boosting local apparel sales by 8–10%. Live demo: https://client-plum-ten.vercel.app | Project repository available for review: https://github.com/deepanmpc/shirt-customizer
- Built interactive 3D shirt preview functionality using Three.js, working directly with the local apparel manufacturer's team to align the customizer with their product specifications and production requirements
- Resolved complex AI integration challenges while iterating on the platform, and received positive in-shop customer feedback on the ease of use of the customizer tool for buyers

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 – Generative AI Professional at Oracle University

Oct 2025 – Oct 2027

Oracle AI Vector Search Professional at Oracle University

Oct 2025 – Oct 2027

SKILLS

Languages: Python, Java, JavaScript, C/C++

Web & Frontend: React, Three.js, TailwindCSS, HTML/CSS, REST APIs

Backend & DevOps: Spring Boot, FastAPI, Docker

Databases: SQL, PostgreSQL, ChromaDB

Tools & Cloud: Git, GitHub, Linux, AWS (basics), OpenAI API, PyTorch, WebRTC, Faster-Whisper, Ollama, Sentence Transformers, Vercel

Soft Skills: Problem Solving, Team Collaboration, Communication, Adaptability, Time Management

PROJECTS

LaRa – Real-Time AI Application Platform

Jan 2026 – Present

- Tech: Python, FastAPI, WebRTC VAD, Faster-Whisper, Docker, Microservices
- Built for **neurodiverse and specially-abled children** — a societal impact platform for accessible, therapeutic AI interaction
- Awarded **3rd Place at RAMPAGE V26** — National Level Hackathon conducted at KL-H
- Designed production-grade microservices with FastAPI and containerised with Docker; integrated real-time speech pipeline (Faster-Whisper + WebRTC VAD) with modular session state and analytics modules

ResumeAnalyse – AI Resume Matching System (RAG Architecture)

Dec 2025 – Feb 2026

- Tech: Sentence Transformers, ChromaDB, Ollama, Mistral LLM, React, TypeScript
- Implemented semantic similarity matching for job-resume ranking achieving over 90% relevance using embedding-based search
- Built a React dashboard supporting resume uploads and AI chat analysis; reduced recruiter analysis time by 50% through automated insights

ISL Sign Language Recognition – CNN Model

Aug 2025 – Nov 2025

- Tech: PyTorch, CNNs, Python, NVIDIA A100 / H100 / T4
- Trained CNN model on 42,000 gesture images achieving 99.8% validation accuracy and 90% test accuracy
- Implemented real-time inference pipeline at 45 FPS on edge devices; designed scalable architecture supporting 1M+ samples on NVIDIA A100, H100, T4 GPUs

CODING PROFILES

LeetCode: https://leetcode.com/u/kl2300032731/ **CodeChef:** https://www.codechef.com/users/klu2300032731